Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Mid-Term Examination, July-December-2023

Program: B. Sc. Engg.(CSE)

Session : **2020-21**

Course Code : CIT-323 | Course Title : Simulation and Modeling

Full Time: 1 Hour | Full Marks: 15

Answer all the questions

1. Distinguish between analytical solution and simulation. 02
2. Explain the problem statement of simulation of a single server queuing system. 05
3. A drunkard moved from a point to a destination and takes step in four directions, backward, forward, right, and left. The probabilities associated with these are 15%, 40%, 20% and 25% respectively. The distances covered to backward, forward, right, and left are 50 cm, 90 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales. [USE Random Number 67, 23, 01, 95, 84, 56, 77, 76, 18, 82, 45, 29, 68, 21, 15, 86, 00, 89, 45] 67 04
4. Define Monte-Carlo method. Solve the integrals $I = \int_1^5 \frac{x^4}{3} dx$ by monte-carlo method. [USE Random Number 48, 36, 44, 41, 46, 18, 32, 41, 23, 23 for X coordinate and Random Number .52, .18, .30, .88, .21, .25, .57, .82, .9, .63 for Y coordinate, you have a right to do scaling, if required] 04

Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2023
Course Code: CCE-323 Course Title: Optical Fiber Communication
Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1 a) An 8 km length of fiber with a mean optical power launch of 120 μW will have a mean optical power of 3 μW at the fiber output. Assess: 4

i. If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber;
ii. the fiber's signal attenuation per kilometer.
iii. the total signal attenuation for a 20 km optical link made of the same fiber with 2 km-separated splices that each provide a 2 dB attenuation;
iv. the numerical input/output power ratio in (c).

b) A typical relative refractive index difference for an optical fiber designed for long distance transmission is 1%. Estimate the NA and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core–cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber. 3

2 a) Compare and contrast, using suitable diagrams, the outside vapor-phase oxidation (OVPO) process and the modified chemical vapor deposition (MCVD) technique for the preparation of low-loss optical fibers. 6

b) Briefly describe the major reasons for the cabling of optical fibers which are to be placed in a field environment. 2

Patuakhali Science and Technology University
Department of Computer Science and Information Technology

6th Semester (Level-3, Semester-II), **Midterm Examination** of **B.Sc. Engg.(CSE)**, July-Decem./2023, Session: 2020-21

Course Code: CIT-321 Course Title: Operating System

Full Marks: 15 Duration: 1 hour

1.	What are the three main purposes of an operating system? Some early computers protected the operating system by placing it in a memory partition that could not be modified by either the user job or the operating system itself. Describe two difficulties that you think could arise with such a scheme.	04
2.	How do clustered systems differ from multiprocessor systems? What is required for two machines belonging to a cluster to cooperate to provide a highly available service?	03
3.	What is critical-section problem? Explain the requirements which must be satisfied to solve the critical-section problem. Review the limitations of semaphore solution to the critical section problem.	04
4.	Explain the strategy, high-level synchronization construct-the monitor type, for dealing with such errors.	04

Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2023
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.0   Full Marks: 15   Duration: 60 Minutes

1	a)	Differentiate between standard I/O and memory mapped I/O.	2
	b)	What are the main types of interrupt? Describe with example.	2
	c)	Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is 9000H.	3.5
2	a)	Explain the different communication models used in IoT. How do these models impact data transmission, scalability, and energy efficiency in IoT systems?	3.5
	b)	What does it mean by Internet of Underground Things and Internet of Underwater Things?	2
	c)	The Internet of Things (IoT) is often described in three key dimensions. Explain each dimension with relevant examples. How do these dimensions collectively enable a seamless IoT ecosystem?	2

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Midterm Examination, July-December-2023

Program: B. Sc. Engg.(CSE)
Session : 2020-21

Course Code : CIT-324 | Course Title : Simulation and Modeling Sessional

Full Time: 1 Hours | Full Marks: 15

Answer the marked question

SET-A

✓ 1. A six-sided die rolled and produces random numbers 1 to 6. Simulate the Gambling Game with six-sided die rolled in odd trail is treated as H and even trail as T. When the differences between H and T will 20 game over. It costs 1 dollar in each trail. Use Monte Carlo Method and analyze profit-loss. 08

2. A flying object moves randomly. It moves L, R, F and B in 25%, 35%, 30% and 10% probability. It also has 80% upward and 20% downward tendency. Identify the location of flying object after 50 epochs by Monte Carlo Method. 08

SET-B

3. Write a program that performs a simple M/M/1 queue simulation. This program requires parameters for the Mean Inter Arrival time of customers, Mean Service time as well as the maximum number of customers. The simulation is started with a single-server queue with a FIFO queuing discipline. For M/M/1 queue, the customer inter-arrival time and the service time are both exponentially distributed. This simulation shows the Average delay in queue, Average number in queue, Server utilization, and Time simulation ended. 07

Patuakhali Science & Technology University (PSTU)

Department of Computer Science and Information Technology(CSIT)

Mid Term Examination: July-December 2023

Course Code: CIT 322 | Course Title: Operating Systems Sessional

Session: 2020-21, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 15

[Answer the marked questions]

SET-A (Any one from 1 to 3)

1. Implement Peterson's solution for ensuring synchronization. 07
2. Implement mutex locks to ensure synchronization.
3. Semaphore implementation to solve busy waiting problem.

SET-B(One option from 4-5; 6 is mandatory)

✓ 4. Write a code to demonstrate how to create and start Python's/Java threads using the threading module. It runs a function print_name that prints a name along with some arguments. This example creates two threads, starts them using the start() method, and waits for them to complete using the join method. 04

5. Write a code to demonstrate how to use Python's/Java threading module to calculate the square and cube of a number concurrently. Two threads, t1 and t2 , are created to perform these calculations. They are started, and their results are printed in parallel before the program prints "Done!" when both threads have finished. Threading is used to achieve parallelism and improve program performance when dealing with computationally intensive tasks. 04

6. 04

odd

I. Displays information about files in the current directory.
II. Create a directory with own student name in which create a file after this create also directory on home directory. Copy the previous created file on home directory.
III. To navigate between different folders.
IV. Removes empty directories from the directory lists.
V. In file copy/cut and paste
VI. download files from the internet.
VII. Displays the current users name
VIII. View Calendar in terminal
IX. Check the details of the file system
X. Check the lines, word count, and characters in a file using different options

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), Jul-Dec 2023

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. (a) Why op-amps are used? Show the pin number of 741C op-amp. 2+3
   (b) Write down the working principle of inverting and non-inverting amplifier with circuit diagram. 6
   (c) Show the difference between close loop gain and open loop gain. 3

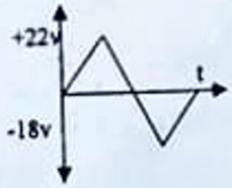
2. (a). Draw the circuits of subtractor and integrator. Write down the working principles of those circuits. 7
   (b) Why filter is used? Show the frequency response of low pass filter and high pass filter. 4
   c) Design a low pass filter. Find the cutoff frequency and voltage gain of the designed filter if R= 12 KΩ and C=0.02μF. 3

3. (a) How transistor is used to design inverter? Calculate $I_{B,sat}$ and $I_{C,sat}$ for your designed inverter if V_{cc} is 8 Volt, R_B is 7 kΩ, R_c is 3 kΩ and h_{FE} is 25. 5
   (b) Draw the circuit of CMOS Inverter. Write down the working principle of CMOS inverter 5
   c) How power dissipation and figure of merit are calculated in digital ICs? 4

4. (a). Explain how D/A converter works. Find the analog out voltage of 5-bit D/A converter for all possible inputs. Assume K=1. 6
   (b) Design a 3 bit parallel comparator A/D converter. Explain how it works. 5
   (c) How quantization error is reduced in A/D converter? 3

5. a) Design a 3 input RTL NOR gate and explain how it works. Calculate noise margin if h_{FE}=25; N=5; R_C= 540Ω; R_B= 660Ω; and V_{cc}= 4.2V. 6
   b) Draw a circuit of 3-input TTL NAND Gate and write down the working principle of TTL NAND Gate. 5
   c) Show the comparison between DTL and TTL. 3

6. a) Show the applications of clipper and clamper in electronic circuits. 4
   a) Analyse the clipper network using ideal diode for given input and circuits Fig. 6(a) and Fig. 6(b). Determine the output waveform of the given network. 6



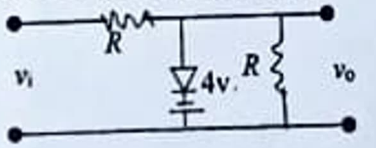


Fig. 6(a)

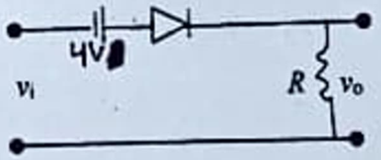


Fig. 6(b)

   b) Design a clamper with ideal diode to perform the given input and output. 4

Patuakhali Science and Technology University
Department of Computer Science and Information Technology

6th Semester (Level-3, Semester-II), Final Examination of B.Sc. Engg.(CSE), July-December/2023, Session: 2020-21

Course Code: CIT-323 **Course Title:** Simulation and Modeling

Full Marks: 70 **Credit hour:** 3.00 **Duration:** 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions.

1. a. Define simulation with example. Demonstrate different ways in which a system might be studied. Explain the next-event time-advance approach illustrated for the single-server queueing system. 05

   b. Consider a single-server queuing system, there are several IID random variables interarrival times, service times, and customer delays. Describe the problem statement of the single-server queuing system. 05

   c. What is an inventory system? Formulate a simulation of an inventory system in which many of the elements are representative of those found in the actual inventory system. 04

2. a. Define random variable. How to compute variance, covariance, and correlation for simulation study? Why stochastic process is used to simulate the output data. Illustrate correlation function of the process of delays in queue D1, D2, ...for the M/M/1 queue. 05

   b. Define verification, validation, and credibility in simulation model. Illustrate the timing relationship of validation, verification, and establishing credibility. 05

   c. Describe the eight methods for debugging a simulation model's computer program. 04

3. a. Write down the properties of the location, scale, and shape parameters of continuous distributions. State the relevant information (application, density function, parameters, and density curve) for the simulation modeling application of the normal continuous distribution. 05

   b. Many random number generators in use today are linear congruential generators, LCGs. Discuss how to obtain the desired random numbers from LCGs? What is the inverse transform function? Let X have an exponential distribution with a mean β. Generate the desired random variate using the inverse transform function. 06

   c. What is the necessity of statistical analysis with simulation output data? Describe and illustrate the transient and steady-state behaviors of a stochastic process. 03

4. a. Define computer simulation models. Describe the steps in simulation study. 02

   b. A machine center has three identical bearings, which fail according to the following probability distribution in table A. The present maintenance policy is to change a bearing as and when it fails. When bearing fails machine center stops, a technician is called to replace the failed bearing with a new one. The time between the failure of the bearing and reporting of the technician (delay time) is random and is distributed as table B. 10

Table: A	
Bearing Life(Hours)	***Probability***
1200	0.10
1400	0.25
1600	0.30
1800	0.25
2000	0.10

Table: B	
Delay Time(Minutes)	***Probability***
2	0.3
5	0.5
8	0.2

   It takes 15 minutes to change one bearing, 25 minutes to change two bearing and 35 minutes to change all three bearings. Downtime of the system costs $5 per minutes, direct on job costs of the technician is $30 per hour and the cost of bearing is $30. The maintenance department is interested in evaluating an alternative policy of replacing all the three bearings, whenever a bearing fails. Simulate the present and proposed policy for 10000 hours and choose the better policy.

c. What is normally distributed random numbers? Describe the properties and applications of normally distributed random numbers. 02

5. a. Write down the qualities of an efficient random number generator. Use mixed congruential method to generate the following sequence of random numbers. A sequence of 10 two digit random numbers such that $r_{n+1} = (19r_n + 1)\ modulo 23$;Take $r_0 = 1$. [Take seed values yourself] 03

b. Define Monte-Carlo method. Solve the integrals $I = \int_1^5 \frac{x^3}{3} dx$ by monte-carlo method. [USE Random Number 48, 36, 44, 41, 46, 18, 32, 41, 23, 23 for X coordinate and Random Number .52, .18, .30, .88, .21, .25, .57, .82, .9, .63 for Y coordinate, you have a right to do scaling, if required] 04

c. There are a number of moving objects, which chase each other in a serial order like A chases B, B chases C, and C chases D etc. Each object moves towards its target unmindful of the fact that it itself is being targeted by some other object. Depending upon the original location, and speed of motion of the object, each object will take its own time to hit its target. As soon as hit occurs, the chase ends. The initial location of the objects A, B, C, and D are (0,0), (0,10), (0,20), (0,30) respectively and their velocities are 30km/hr, 25km/hr, 20 km/hr and 15km/hr respectively. Object D moves towards a fixed at (30,50), while C moves always in a direction pointing towards D, the object B moves always pointing towards C and object A always pointing towards B. It can be assumed that when the distance between two objects less than 0.005 hit is taken to occur. Identify, which object will hit the target at first and in what time? 07

6 a. What are the differences between true and pseudo random numbers? Describe the procedure to physically generate random number at the interval [0, 1] with two-digit accuracy. 02

b. A sequence of 10,000 five digit random numbers has been generated, and an analysis of numbers indicate that there 3075 numbers having five different digits,4935 having a pair,1135 having two pairs, 695 having three of a kind,105 having full house, 54 having four of a kind and one having all five of a kind. Use Poker test to determine if these random numbers are independent at $\alpha = 0.01$. Note that for $\alpha = 0.01$ and N=06 the critical value is 16.8. 06

c. An ammunition depot, of rectangular shape measuring 500 m of length and 350m of width is under attack by a squadron of bombers. In each sortie 10 bombers drop bombs, one each, on the ammunition depot. If the bomb lands anywhere in the marked area, it is a hit, otherwise it is a miss. All the bombers aim at the center of the area. The point of strike is assumed to be normally distributed around the aiming point with a standard deviation of 500 m in x-direction and 350m in the y-direction. Simulate operation of bombing operation for 20 strikes and determine the percentage number of strikes, which are on the target. 06

Use Normal Random Number:

For X direction – [.23, -1.16, 0.39, -1.90, -0.78, -0.02, -0.40, -0.66, 1.14, 0.07, 0.53, 0.03, 1.19,0.11, 0.16, -0.82, 0.42, -0.89, 0.49, -0.21]

For Y direction- [0.24, -0.02, 0.64, -1.04, 0.68, -0.47, -0.75, -0.44, 1.21, -0.08, -1.56, 1.49, -1.19, -1.86, 1.21, 0.75, -1.50, 0.19, -1.44, 0.65]

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of computer Science and Information Technology

6th Semester (Level-3, Semester-II), **Final Examination** of B.Sc. Engg. CSE, July-December/2023

Session: 2020-21

Course Code: CIT-321 Course Title: Operating System

Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks]

Answer any 5 of the following questions.

1. (a) Describe the general organization of a computer system and the role of interrupts. Describe the components in a modern multiprocessor computer system. 3

   (b) Illustrate the transition from user mode to kernel mode. How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security)? 4

   (c) Compare and contrast monolithic, layered, microkernel, modular, and hybrid strategies for designing operating systems. 3

   (d) Identify several advantages and several disadvantages of open-source operating systems. Identify the types of people who would find each aspect to be an advantage or a disadvantage. 4

2. (a) Identify the separate components of a process and illustrate how they are represented and scheduled in an operating system. 3

   (b) Define remote procedure calls. Describe and contrast interprocess communication using shared memory and message passing. 4

   (c) Identify the basic components of a thread, and contrast threads and processes. Describe the major benefits and significant challenges of designing multithreaded processes. 3

   (d) What resources are used when a thread is created? How do they differ from those used when a process is created? Illustrate different approaches to implicit threading, including thread pools, fork-join, and Grand Central Dispatch. 4

3. (a) Describe various CPU scheduling algorithms. Assess CPU scheduling algorithms based on scheduling criteria. 4

   (b) Explain the difference between preemptive and nonpreemptive scheduling. Explain the issues related to multiprocessor and multicore scheduling. 4

   (c) The following processes are being scheduled using a preemptive, priority-based, round-robin scheduling algorithm. 6

Process	Burst Time	Priority	Arrival
P1	15	8	0
P2	20	3	0
P3	20	4	20
P4	20	4	25
P5	5	5	45
P6	15	5	55

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. The scheduler will execute the highest priority process. For processes with the same priority, a round-robin scheduler will be used with a time quantum of 10 units. If a process is preempted by a higher priority process, the preempted process is placed at the end of the queue.

i. Show the scheduling order of the processes using a Gantt chart.
ii. What is the turnaround time for each process?
iii. What is the waiting time for each process?

4. (a) What causes data inconsistency during concurrent access? What is the critical-section problem? Explain the requirements that must be met to solve the critical section problem. 4

(b) Mention the limitation of Peterson's solution for addressing the critical section problem. State the solution to the critical section problem using mutex locks. Review the limitations of semaphore solution to the critical section problem. Explain the strategy, high-level synchronization construct- the monitor type, for dealing with such errors. 7

(c) What is spinlock? State the effect of liveness failure in the process execution life cycle. 3

5. (a) What is a system model? Define deadlock with a multithreaded application example. Point out the necessary conditions, which hold simultaneously in a system, for a deadlock situation. 4

(b) List the methods for handling deadlock. Mention the possible side effects of preventing deadlock. What is deadlock avoidance? Illustrate safe, unsafe, and deadlocked state spaces with an example. 5
2

(c) How to detect deadlock in a system? Illustrate and explain the data structures, which must be maintained, the process, and an example of the deadlock detection algorithm with several instances of each resource type. 5
1

6. (a) Define and demonstrate the dining-philosophers problem as a synchronization problem. State the semaphore and monitor solutions of the dining-philosophers problem. 3

(b) Distinguish logical versus physical address space. What is a memory management unit? Explain the paging process, a memory-management scheme that permits a process's physical address space to be noncontiguous. 4

(c) What is the purpose of system calls? What system calls have to be executed by a command interpreter or shell in order to start a new process on a UNIX system? 3

(d) Why do some systems store the operating system in firmware, while others store it on disk? Describe three general methods for passing parameters to the operating system. 4
2

Dept. of Computer and Communication Engineering
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Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: III Semester: II Session: 2020-2021

Course Code	Course Title	July-December 2023	Credit: 03
CCE-321	Computer Peripheral and Interfacing		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Which design features are essential in WSNs to optimize data collection, communication, and energy efficiency? 3 2

[B.] Explain the structure of a potentiometer sensor used for measurement of linear displacement with electric circuit diagram and proper equation. List the applications of potentiometer sensor in/around your home and office/university. 4

[C.] Write down the primary objective and idea of Timeout-MAC in wireless sensor networks. How does T-MAC differ from S-MAC in terms of energy efficiency and what mechanism does T-MAC use to reduce idle listening in sensor nodes? 4

[D.] Define proximity switches. Sketch the configurations of different contact-type proximity switch being used in manufacturing automation. Explain in brief two applications of "Reed switch". 3 2

2 [A.] 'Bellows are more sensitive than capsules in Fluid pressure sensor'. State true or false and justify your answer. 3

[B.] How does a capacitive sensor operate to measure linear displacements, and what are the typical applications where such non-contact sensors are preferred over other types of displacement sensors? Explain all with appropriate diagram. 4

[C.] Write down about Hidden Terminal Problem and Exposed Terminal Problem for the CSMA MAC-Protocol in wireless sensor network. Explain with suitable figure. 4

[D] Briefly expailn the following IoT data protocols with their operation. 3

i. MQTT

ii. M2M

3 [A.] What do you mean by nonlinearity error? How it is different than Hysteresis error? 2

[B.] How does a Hall effect sensor detect the presence and strength of a magnetic field? Explain the principle of working of Hall effect sensor. 3

[C.] What is a LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. 5 5

[D.] How does **Slotted ALOHA** improve upon the basic ALOHA protocol in terms of efficiency? Explain with frames transition in a pure ALOHA and a **Slotted ALOHA** network. 4

4 [A.] Draw the architectural diagram of 8279 programmable keyboard/display controller. How does 8279 keyboard work? 5

[B.] Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is A000H. Draw a neat logic diagram of your implementation. 4

[C.] Given a memory with a 14-bit address and an 8-bit word size. 3

i. How many bytes can be stored in this memory?

ii. If this memory were constructed from 1K x 1 RAMS, how many memory chips would be required?

iii. How many bits would be used for chip select?

[D.] Define the three types of I/O. Identify each as either microprocessor-initiated or device-initiated. 2

5 [A.] Show the software keyboard interfacing process through a flow chart. Describe each step for a 4x4 keyboard. 5

[B.] Consider the hardware schematic shown in **Figure 01.** 4

i. Determine the address map of this system. Note: $\overline{MEMR} = 0$ for read, $\overline{MEMR} = 1$ for write, $M/\overline{IO} = 0$ for I/O and $M/\overline{IO} = 1$ for memory.

ii. Is there any possibility of bus conflict in this organization? Clearly justify your answer.

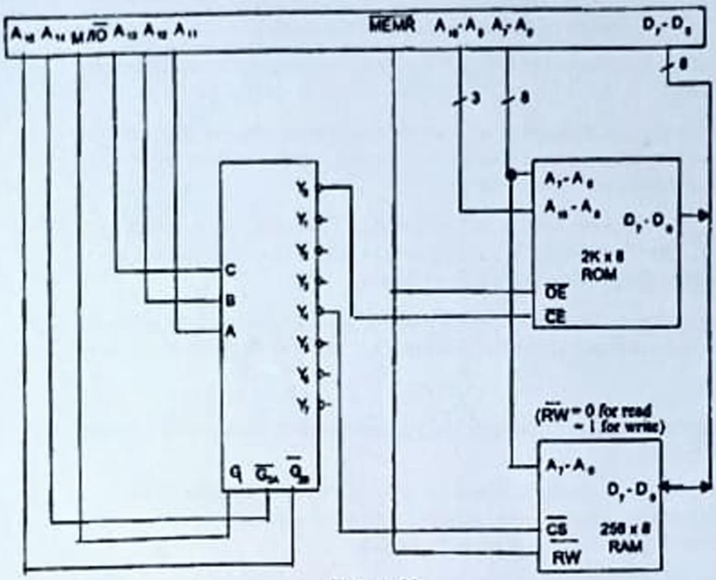

Figure: 01

[C.] Summarize the basic difference between maskable and nonmaskable interrupts. Describe how power failure interrupt is normally handled. 3

[D.] i. What is fluorescence? 2 1

ii. If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?

6 [A.] The block diagram of a 512 x 8 RAM chip is shown in **Figure 02.** In this arrangement the memory chip is enabled only when $\overline{CS1}$ = L and CS2 = H. Design a 1K x 8 RAM system using this chip as the building block. Draw a neat logic diagram of your implementation. Assume that the microprocessor can directly address 64K with a $R/\overline{W}$ and eight data pins. Using linear decoding and don't-care conditions as 1's, determine the memory map in hexadecimal. 4

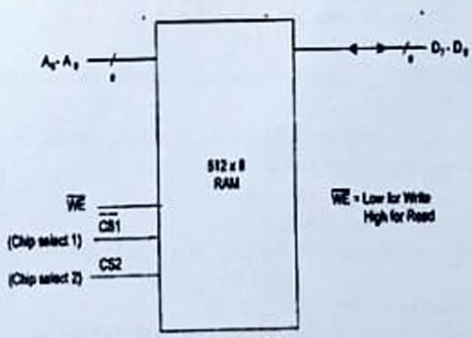

Figure: 02

[B.] Serialize the steps associated in turning on or off a pixel in LCD monitor. 3

[C.] List the methods used in main memory array design. What are the advantages and disadvantages of each? 3 2

[D.] Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port. 2

[E.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer. 2

Dept. of Computer and Communication Engineering
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Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2020-21

Course Code	Course Title	July-December 2023	Credit: 03
CCE-323	Optical Fiber Communication		Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] i. A long single-mode optical fiber has an attenuation of 0.5 dB km^{-1} when operating at a wavelength of 1.3 μm. The fiber core diameter is 6 μm and the laser source bandwidth is 600 MHz. Compare the threshold optical powers for stimulated Brillouin and Raman scattering within the fiber at the wavelength specified. **4**

ii. Describe different advantages of optical fiber communication. **3** 2

[B.] i. A 10 km length of fiber with a mean optical power launch of 120 μW will have a mean optical power of 4 μW at the fiber output. **4**

Assess:

a) If no connectors or splices are present, the total signal attenuation or loss in decibels via the fiber;
b) the fiber's signal attenuation per kilometer,
c) the total signal attenuation for a 20 km optical link made of the same fibe r with 2 km-separated splices that each provide a 2 dB attenuation;
d) the numerical input/output power ratio in (c).

ii. What is optical fiber? Show its typical configuration with necessary diagram. **3**

2 [A.] What are the sources of fiber optic losses in silica? Provide a comprehensive explanation of the intrinsic absorption loss mechanisms, including relevant examples. **7**

[B.] Two step index fibers exhibit the following parameters: **7** 3

a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm;
b) a 8 μm core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.3% and an operating wavelength of 1.55 μm.

Estimate the critical radius of curvature at which large bending losses occur in both cases.

3 [A.] What sets an optical fiber communication system apart from a conventional communication system? Illustrate the optical fiber communication system using an appropriate block diagram. **7**

[B.] Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7\times10^{-11} m^2N^{-1}$. The refractive index and the photo elastic coefficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm. Boltzmann's constant is $1.381\times10^{-23} JK^{-1}$. **7**

4 [A.] What are the direct and indirect Band Gap Semiconductors? **3**

[B.] Explain the working principle of Expanded Beam Connectors used in optical fiber communication. Discuss their advantages and disadvantages compared to physical contact fiber connectors. **4**

[C.] Discuss the mechanism of optical feedback to provide oscillation and hence amplification within the laser. Indicate how this provides a distinctive spectral output from the device. 4

[D.] The radiative and non-radiative recombination lifetimes of the minority carriers in the active region of a double-heterojunction LED are 60 ns and 100 ns respectively. Determine the total carrier recombination lifetime and the power internally generated within the device when the peak emission wavelength is 0.87 μm at a drive current of 40 mA. 3

5 [A.] Describe, two common techniques used for the mechanical splicing of optical fibers with the suitable diagrams. 4

[B.] Explain the operation of a substrate-entry InGaAs p-i-n photodiode by: 4

i. Drawing and labeling its layer structure.

ii. Sketching the energy band diagram and discussing the effects of the heterojunction on charge trapping and carrier dynamics.

[C.] Two multimode step index fibers have numerical apertures of 0.2 and 0.4, respectively, and both have the same core refractive index of 1.48. Estimate the insertion loss at a joint in each fiber caused by a 5° angular misalignment of the fiber core axes. It may be assumed that the medium between the fibers is air. 3

[D.] What is the Fermi–Dirac distribution, and how does it describe the probability of electron occupancy in energy states of a semiconductor? 3

6 [A.] With the aid of simple sketches outline the major categories of multiport optical fiber coupler. Describe two common methods used in the fabrication of three- and four-port fiber couplers. 5

[B.] A photodiode has a quantum efficiency of 65% when photons of energy 1.5 × 10−19 J are incident upon it. 3

(i) At what wavelength is the photodiode operating?

(ii) Calculate the incident optical power required to obtain a photocurrent of 2.5 μA when the photodiode is operating as described above.

[C.] What is the main difference between LED and Laser? 2

[D.] How does the double-heterojunction structure enhance the performance of an LED? Support your answer with a labeled diagram of the layer structure under forward bias and the corresponding energy band diagram. 4

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology(CSIT)
Final Examination: July-December 2023
Course Code: CIT 322 | Course Title: Operating Systems Sessional
Session: 2020-21, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 70

[Answer the marked questions]

SET-A

1. The following processes are being scheduled using a preemptive, priority-based scheduling algorithm.[USE CONCURRENT PROCESS] 15

Process	Priority	Burst Time	Arrival Time
P1	8	15	0
P2	3	20	0
P3	4	20	20
P4	4	20	25
P5	5	5	45
P6	5	15	55

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. The scheduler will execute the highest priority process. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

i. What is the turnaround time for each process?
ii. What is the waiting time for each process?

2. Show the difference between preemptive and non-preemptive scheduling. Consider the following set of processes, with the length of the CPU burst given in milliseconds:[USE CONCURRENT PROCESS] 15

Process	Burst Time	Priority
P1	5	4
P2	3	1
P3	1	2
P4	7	2
P5	4	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

i. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).

	ii. What is the turnaround time of each process for each of the scheduling algorithms in part a? iii. What is the waiting time of each process for each of these scheduling algorithms? Which of the algorithms results in the minimum average waiting time (over all processes)?	
3.	Create some concurrent process and apply round-robin CPU scheduling algorithm on them.	15
4.	Create some concurrent process and apply priority CPU scheduling algorithm on them.	15
x	Create some concurrent process and apply First Come First Serve (FCFS) CPU scheduling algorithm on them.	15
5.	Create some concurrent process and apply Shortest job first (Non- Preemption) CPU scheduling algorithm on them	15
✓ 6.	Create some concurrent process and apply SJF CPU scheduling algorithm on them	15
7.	Create some concurrent process and apply SRT CPU scheduling algorithm on them	15
8.	Create a client server project using Windows Server OS. (Project)	10
	SET -B	
B-1 9.	Implement Peterson's Algorithm for the two processes using shared memory so that they are mutually exclusive. The solution should be free from synchronization problems.	15
10.	Implement the Dining Philosopher problem where K philosophers are seated around a circular table with one chopstick between each pair of philosophers.	10
B-2 11.	Write a program to implement the banker's algorithm for deadlock avoidance.	15
	A project on the given problem.	10
13	Viva-Voce	20
12.	Reader - Writer problem	10

Patuakhali Science & Technology University (PSTU)

Department of Computer Science and Information Technology(CSIT)

Final Examination: July-December 2023

Course Code: CIT 324 | Course Title: Simulation and Modeling Sessional

Session: 2020-21, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 70

[Answer the marked questions]

Section A

1. You are tasked with generating pseudo-random numbers using the Mid-Square Method and using those numbers to perform the Kolmogorov-Smirnov (K-S) test for uniformity. Start with a 4-digit seed (e.g., 5731). 10
2. You are tasked with evaluating the quality of a random number generator using the Multiplicative Congruential Method (MCM) and verifying its uniformity using the Chi-Square goodness-of-fit test. 10
3. You are tasked with analyzing the randomness of a sequence of numbers generated using the Additive Congruential Method (ACM). To evaluate the independence of these random numbers, you will implement a Chi-Square test for autocorrelation. 10
4. Implement the Poker Test to assess the randomness of numbers generated by a Multiplicative Congruential Generator (MCG). 10

Section B

5. You are tasked with simulating a mobile robot that follows a predefined path using the Pure Pursuit algorithm. The robot should start at an initial position and navigate towards a sequence of waypoints using the pure pursuit strategy. Your solution must include a real-time animation of the robot moving along the path. 20

6. You are tasked with simulating the chemical reaction between hydrogen (H_2) and oxygen (O_2) to form water (H_2O). The balanced chemical reaction is: $2H_2 + O_2 \rightarrow 2H_2O$ 20
   Given an initial amount of hydrogen (h2_molecules) and oxygen (o2_molecules), simulate the reaction until no more water can be formed (i.e., one or both reactants are exhausted). Then, determine the equilibrium state — the remaining molecules of hydrogen, oxygen, and water. Your task is to:
   1. Simulate the reaction in Python.
   2. Output the number of remaining hydrogen, oxygen, and the amount of water formed when the reaction stops (reaches equilibrium).

7. In the Serial Chase Problem, multiple entities (such as agents, cars, or robots) are positioned on a circular path. Each entity is programmed to chase the next one in the sequence, forming a closed loop. Each chaser always moves directly toward its target at a constant speed. 20
   Task:
   Write a simulation program to model the Serial Chase Problem for n entities placed evenly on a 2D circular path. Your simulation should:
   1. Initialize n entities equally spaced on a unit circle.
   2. Each entity should continuously move toward the next one (e.g., entity 1 chases entity 2, entity 2 chases entity 3, ..., and entity n chases entity 1).
   3. Update positions using a small time step Δt and simulate until all entities converge to a single point or until a maximum number of iterations is reached.
   4. Display or output the trajectories of each entity.
   5. (Optional) Visualize the simulation using a plotting library.

8. A machine center has three identical bearings, which fail according to the following probability distribution in table A. The present maintenance policy is to change a bearing as and when it fails. When bearing fails machine center stops, a technician is called to replace the failed bearing with a new one. The time between the failure of the bearing and reporting of the technician (delay time) is random and is distributed as table B. 20

Table: A

Bearing Life(Hours)	*Probability*
1200	0.10
1400	0.25
1600	0.30
1800	0.25
2000	0.10

Table: B

Delay Time(Minutes)	*Probability*
2	0.3
5	0.5
8	0.2

It takes 15 minutes to change one bearing, 25 minutes to change two bearing and 35 minutes to change all three bearings. Downtime of the system costs $ 5 per minutes, direct on job costs of the technician is $ 30 per hour and the cost of bearing is $ 30. The maintenance department is interested in evaluating an alternative policy of replacing all the three bearings, whenever a bearing fails. Simulate the present and proposed policy for 10000 hours and choose the better policy.

Section C

10

09. Write a program that performs a simple M/M/1 queue simulation. This program requires parameters for the Mean Inter Arrival time of customers, the Mean Service time as well as the maximum number of customers. The simulation starts with a single-server queue with a FIFO queuing discipline. For M/M/1 queue, the customer inter-arrival time and the service time are both exponentially distributed. This simulation shows Average delay in queue, Average number in queue, Server utilization, and Time simulation ended.

10. LCG is considered one of the basic yet best methods to generate Pseudo-Random Numbers. Write a program that generates a random number using the LCG.

Section D

10

11. There were a few waves in the COVID-19 pandemic. Let a wave occur every 100 days in Bangladesh, on average. After a wave occurs, find the probability using the Exponential distribution that it will take more than 120 days for the next wave to occur. Simulate several Exponential distributions using rate parameters 0.5, 1.0, 2.0, and 4.0.

12. There are several general approaches to generating a univariate Random Variable from the distribution function. Write a program that generates the desired number of random variates using the inverse transform function.

20

13. Viva-Voce

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University

Final Examination of B. Sc. Engineering in CSE Level: III Semester: II Session: 2020-2021

Course Code	**Course Title**	**July-December** 2023	**Credit**: 1.50
CCE-322	Computer Peripheral and Interfacing sessional		**Time**: 03 Hr
			Marks: 70

1.	**Viva**	**20**
2.	**Write down project title, objectives.**	**5**
3.	**Circuit diagram/ flow chart / equipment**	**5**
4.	**Project**	**30**
5.	**Report**	**5**
6.	**Continuous assessment**	**5**

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